

HASH TABLE ASSIGNMENT RESULTS

Assignment Details

- Hash Function: $h(k,i) = (k + i^3) \bmod 10$
- Keys to Insert: 15, 27, 5, 12, 35
- Table Size: 10
- Method: Open Addressing with Cubic Probing

Initial Hash Table

Index	0	1	2	3	4	5	6	7	8	9
Value	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1

Step-by-Step Insertion Process

Step 1: Inserting Key 15

Calculation: $h(15,0) = (15 + 0^3) \bmod 10 = 15 \bmod 10 = 5$

Result: No collision → Insert at index 5

Index	0	1	2	3	4	5	6	7	8	9
Value	-1	-1	-1	-1	-1	15	-1	-1	-1	-1

Step 2: Inserting Key 27

Calculation: $h(27,0) = (27 + 0^3) \bmod 10 = 27 \bmod 10 = 7$

Result: No collision → Insert at index 7

Index	0	1	2	3	4	5	6	7	8	9
Value	-1	-1	-1	-1	-1	15	-1	27	-1	-1

Step 3: Inserting Key 5

Calculation: $h(5,0) = 5 \rightarrow$ COLLISION!

$h(5,1) = (5+1^3) \bmod 10 = 6$

Result: 1 collision $\rightarrow$ Insert at index 6

Index	0	1	2	3	4	5	6	7	8	9
Value	-1	-1	-1	-1	-1	15	5	27	-1	-1

Step 4: Inserting Key 12

Calculation: $h(12,0) = (12 + 0^3) \bmod 10 = 12 \bmod 10 = 2$

Result: No collision $\rightarrow$ Insert at index 2

Index	0	1	2	3	4	5	6	7	8	9
Value	-1	-1	12	-1	-1	15	5	27	-1	-1

Step 5: Inserting Key 35

Calculation: $h(35,0) = 5 \rightarrow$ COLLISION!

$h(35,1) = 6 \rightarrow$ COLLISION!

$h(35,2) = (35+8) \bmod 10 = 3$

Result: 2 collisions $\rightarrow$ Insert at index 3

Index	0	1	2	3	4	5	6	7	8	9
Value	-1	-1	12	35	-1	15	5	27	-1	-1

Final Results Summary

Key	Final Position	Collisions	Probe Sequence
15	5	0	$h(15,0) = 5$
27	7	0	$h(27,0) = 7$
5	6	1	$h(5,0) = 5 \rightarrow h(5,1) = 6$
12	2	0	$h(12,0) = 2$
35	3	2	$h(35,0) = 5 \rightarrow h(35,1) = 6 \rightarrow h(35,2) = 3$

Assignment Statistics

Metric	Value
Total keys inserted	5
Total collisions encountered	3
Load factor	50% (5/10 slots)
Empty slots remaining	5 (indices: 0, 1, 4, 8, 9)
Maximum probe distance	2 (for key 35)
Average probe distance	0.6